

Complex Analysis – 6

Целые и мероморфные функции: особенности и рост

Домашнее задание

1 сентября 2026 года

Теоретический минимум. Classification of isolated singularities; Riemann's theorem, Casorati–Weierstrass and Picard; Cauchy estimates and Liouville; order and type of an entire function; the Mittag-Leffler expansion and the Weierstrass product; elliptic functions and the vanishing of the sum of residues over a fundamental domain.

1. Prove the Cauchy estimates and deduce that an entire function with $|f(z)| \leq C(1 + |z|)^m$ is a polynomial of degree at most m . Give an entire function that is bounded on every line through the origin but is not constant.
2. (S.T. Yau CSMC 2023) Let f be entire and suppose there is $C > 0$ with $|f(z)| \leq C\sqrt{|z|}|\cos z|$ for all $z \in \mathbb{C}$. Prove that $f \equiv 0$.
3. (S.T. Yau CSMC 2020) Let f and g be entire functions with $f(z)^{2020} + g(z)^{2020} = 1$ for all z . Prove that f and g are constant. (*Подсказка: сначала случай показателя 2, где помогает параметризация; в общем случае нужна теорема Пикара.*)
4. (S.T. Yau CSMC 2019, team) Is there an entire function f with infinitely many zeros such that for every $r \in (0, 1)$ there are constants $A_r, B_r < \infty$ with $|f(z)| \leq A_r e^{B_r |z|^r}$ for all z ?
5. (S.T. Yau CSMC 2014, team) Prove that every bounded analytic function F on the annulus $\{r < |z| < R\}$ can be written as $F(z) = z^\alpha f(z)$, where f is analytic on the disc $|z| < R$ and α is a constant; deduce the removability of the singularity when $r = 0$.

6. Prove the Mittag-Leffler expansion $\pi \cot \pi z = \frac{1}{z} + \sum_{n \neq 0} \left(\frac{1}{z-n} + \frac{1}{n} \right)$ and the partial fraction expansion of $\pi / \sin \pi z$; deduce $\sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}$ by expanding at $z = 0$.
7. Prove the Weierstrass product formula $\sin \pi z = \pi z \prod_{n \geq 1} \left(1 - \frac{z^2}{n^2} \right)$, and deduce Wallis' product and the value $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.
8. Define the order and the type of an entire function, compute them for e^z , $\sin z$, $\cos \sqrt{z}$ and $1/\Gamma(z)$, and prove that a function of order ρ with $n(r)$ zeros in $|z| \leq r$ satisfies $n(r) = O(r^{\rho+\varepsilon})$.
9. Prove that a non-constant doubly periodic meromorphic function has at least two poles in a fundamental parallelogram (counted with multiplicity), that the sum of its residues there is zero, and that it takes every value the same number of times.
10. Prove the big Picard theorem for the example $f(z) = e^{1/z}$ by hand: describe the image of every punctured neighbourhood of the origin, and state precisely which value is omitted.